Business Statistics

Types and Orientation of Data

Datasets used in this presentation and accompanying MS Excel models are simulated by artificial intelligence. While they are realistic, they are not from actual companies or pre-existing data sets.

Types of Data
Overview

- Understanding data
- Structured vs. Unstructured Data
- Cross-Sectional and Time Series Data
- Stacked vs. Unstacked Data
- Big Data

Understanding Data



Understanding data
What data is needed and what data is available?

- The business question comes before consideration of data.
  - “How does customer behavior differ across our regions?”
  - “Did customer behavior change after our marketing campaign?”
- After asking the business question, a good first step is to evaluate existing data to see if it is adequate to answer the question.
- If it is not, then what additional data needs to be collected, and how will it be integrated with the existing data set?
- Because types of data impact the statistical tools and techniques that can be used, understanding the data is important before studying statistics

Understanding data
Collecting and recording the data

- Assume we are collecting data about different store locations
- Each store is a **subject** with an **identifier** (Store A, Store B, etc.)
- Each subject has multiple **characteristics** or facts about that subject
  - Location, sales amounts, number of employees, etc.
- The single value for each of these **characteristics** is an **observation**
  - We “observed” this fact about the subject
  - We “observed” the location of Store A, and we “observed” the sales of Store B
- A data set contains all of the collected observations of the **characteristics** for one or more **subjects**

Types of Data

Types of data
Structured data

- Structured data has a pre-defined format in which the observations for each subject are stored
- Structured data is generally laid out in rows and columns:
  - Each row corresponds to a subject (record)
  - Each column corresponds to a characteristic of the subject (variable)

- This table has 4 records for the subjects:
  - Store A, B, C, and D
- There are 2 characteristics per subject:
  - Store location and its sales for May

Store ID	Location	Sales In May
A	Downtown	42,000
B	Downtown	35,000
C	Airport	58,000
D	Campus	31,000

Types of data
Unstructured data

- Historically data was organized to make record keeping efficient and analysis feasible – hence the structured data format
- This constrained the type and amount of data that was maintained
- In unstructured data, there is no strict row and column format
- It can have varied types of data, not just text and numbers
  - Pictures, emails, videos, notes, descriptions, etc.
- Today there is value in unstructured data and digital storage is cheap, so unstructured data is kept and included within analysis

Types of data
Structured data is generally of two types

- Cross-sectional data
  - Envision a picture – shows many things at once
  - Many subjects, one point in time
  - The subject is what distinguishes one row (record) from another
- Time Series data
  - Envision a video – tracks one thing over time
  - One subject, many points in time
  - The time period when the data occurred is what distinguishes one observation from another

Cross-sectional data example
How did different coffee shops perform last month?

- Different shops, same time
- Varying: locations, sales
- Fixed: time (month of May)

Store ID	Location	Sales In May
A	Downtown	42,000
B	Downtown	35,000
C	Airport	58,000
D	Campus	31,000

- Enables comparison of performance **across** shops at a **fixed point in time**

Time-series data example
How has a shop's performance changed over time?

- Same shop, different times
- Fixed: shop, location
- Varying: time (month of the sales), sales

Month	Sales
Jan	$38,000
Feb	$41,000
Mar	$42,000
Apr	$46,000

- Enables analysis of a **single** shop's performance **over time**

Longitudinal (panel) data example
Combines cross-sectional and time-series data

- Multiple subjects over multiple time periods

Monthly Sales by Region

Region	January	February	March	April	May
North	41,250	38,900	92,410	27,580	105,320
South	34,180	29,760	81,430	22,950	90,615
East	46,500	42,210	88,975	31,640	97,880
West	39,720	33,415	85,260	28,990	93,540

- Enables comparison of trends between multiple subjects

Data Orientation / Format



Data Orientation, Layout
The same data can appear in different layouts

- Here are 3 months of customer and sales data for Stores A and B

Month	Sales A	Customers A	Sales B	Customers B
Jan	42000	820	31000	640
Feb	39000	780	29000	610
Mar	45000	860	34000	690

- This is panel data as it contains data for multiple subjects and multiple periods

StoreID	Sales Jan	Customers Jan	Sales Feb	Customers Feb	Sales Mar	Customers Mar
A	42000	820	39000	780	45000	860
B	31000	640	29000	610	34000	690

- The data can be oriented differently, with different advantages and disadvantages

StoreID	Month	Sales	Customers
A	Jan	42000	820
A	Feb	39000	780
A	Mar	45000	860
B	Jan	31000	640
B	Feb	29000	610
B	Mar	34000	690

Data Orientation, Layout
Data can be laid out in multiple ways

- Report formatting
  - Used for reports and presentations and not easily used for analysis
- Unstacked data layout
  - Data is laid out in a wide table – easy to read
  - Each row includes all of the characteristics for a subject (many columns)
- Stacked data layout
  - Data is laid out in a tall (long) table – easy for analysis
  - Columns include the subject, characteristic, and value for that characteristic (few columns)

Data Orientation
Reports and presentations

- Oriented specifically for reports
- Data is separated out or broken down by one or more characteristics
- Visually emphasizes groupings
- Not formatted for analysis

	Urban		Suburban	
Store Type	Customers	Sales	Customers	Sales
Mall	200	180,000	800	600,000
Outlet	450	200,000	350	70,000

	Month	Sales
North	January	41,250
	February	38,900
	March	92,410
	April	27,580
	May	105,320

Envision additional tables for the regions South, East, and West and subtotals in a separate column or row

Data Orientation
Unstacked (pivoted) data – easy for people to read

StoreID	Store Type	Customer Satisfaction	Store Open Time (AM)	Monthly Sales
101	Urban	High	8	48,500
102	Suburban	Medium	9	39,200
103	Rural	Low	10	27,400
104	Urban	High	8	51,800
105	Suburban	Medium	9	40,100

- Each subject has its own row (store)
- Each characteristic has its own column
- Each observation is in a row-column intersection

Data Orientation
Stacked (unpivoted) data – easy to perform analysis

- Each observation has its own row
- Included in the observation is
  - Subject
  - Characteristic
  - Value
- Orientation used for statistical models, pivot tables, certain charts

StoreID	Characteristic	Value
101	StoreType	Urban
101	CustomerSatisfaction	High
101	StoreOpenTime	8
101	MonthlySales	48,500
102	StoreType	Suburban
102	CustomerSatisfaction	Medium
102	StoreOpenTime	9
102	MonthlySales	39,200

Data Orientation
Stacked data – can have multiple characteristics

- Each observation has its own row
- Included with the observation is
  - Subject — StudentID
  - Characteristic 1 — Term
  - Characteristic 2 — Course
  - Characteristic 3 — Metric
  - Value — Value
    - Value varies depending on metric

StudentID	Term	Course	Metric	Value
2001	Fall	Math	ExamScore	82
2001	Fall	Math	Attendance	0.92
2001	Spring	Stats	ExamScore	88
2001	Spring	Stats	Attendance	0.95

Big Data



Big Data – IT and Analytics Challenge
Much more than just lots of data

- Big data includes sets of data that are too large and complex to be stored and analyzed by traditional data-processing tools
- Digitally generated data creates massive amounts of real-time, structured and unstructured data
- This all can be used to support better business decisions
- However, the IT and analytical requirements to make that a reality are challenging

Big Data – IT and Analytics Challenge
The 5 V's of Big Data

1. **Volume**: large amounts of data
   - Millions of sales records or website clicks
2. **Velocity**: speed at which its generated and processed
   - Real-time credit card transactions
3. **Variety**: different forms of data
   - Structured and unstructured
4. **Veracity**: data quality and reliability
   - Incomplete or biased data can lead to bad decisions
5. **Value**: usefulness of the data
   - **Data only matters if it leads to better decisions**